

Shanto Mariam University of Creative Technology
Department of CSE and CSIT

PROJECT REPORT ON

"Tanzu's Creation"

Developed Using JavaFX

1. Title Page

Project Title: Tanzu's Creation

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Date: 29.07.2025

2. Acknowledgment

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3. Abstract

This project, “**Tanzu’s Creation**”, is a JavaFX-based desktop application developed as part of the course **CSE-3170: Object Oriented Software Development Lab**. The primary purpose of the application is to provide a user-friendly e-commerce platform that facilitates product browsing, order management, and administrative control within a single system. Addressing the common challenges faced by both customers and administrators—such as secure authentication, efficient product management, and streamlined order processing—the application delivers an integrated solution suitable for educational and practical demonstration.

Key features include user registration and login, role-based access (admin and customer), product catalog management, shopping cart functionality, order placement, and password recovery. Administrative users are empowered to manage products, view orders, and monitor registered users through dedicated panels.

The project utilizes a range of modern technologies and frameworks: **Java** as the programming language, **JavaFX** for the graphical user interface, **FXML** for UI layout definition, and **CSS** for styling. The backend relies on a **MySQL** database to manage persistent data including users, products, and orders. This combination of technologies ensures a responsive, maintainable, and scalable application architecture, serving as a comprehensive example of object-oriented software development in practice.

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5. Introduction

Problem Statement

In today's digital era, both businesses and consumers seek efficient, accessible, and user-friendly platforms for product browsing and order management. Many small businesses and local vendors, however, lack the technical resources to set up such systems. This creates a gap where customers have limited access to products and convenient online ordering, while administrators face challenges in managing inventory and sales records.

Purpose of the Application

The purpose of **Tanzu's Creation** is to bridge this gap by providing a comprehensive e-commerce platform that simplifies product management and order processing. The application is designed to facilitate seamless interaction between customers and administrators, offering secure authentication, intuitive navigation, and easy access to all core functionalities.

Background Study

With the rapid growth of e-commerce and digital marketplaces, the demand for scalable and maintainable desktop applications has increased. JavaFX, combined with modern development practices such as MVC architecture, FXML layouts, and MySQL databases, provides a robust framework for building such systems. The adoption of these technologies not only ensures a smooth user experience but also serves as a valuable learning tool for students studying object-oriented software development.

Target Users

The primary target users of this application are:

- **Customers:** Individuals looking to browse, select, and purchase products online in a secure and convenient manner.
- **Administrators:** Business owners or staff responsible for managing product catalogs, processing orders, and overseeing platform users.

6. Literature Review / Existing System

Existing Solutions

There are numerous existing solutions aimed at managing e-commerce platforms for both customers and administrators. Web-based systems like Shopify, WooCommerce, and Magento offer robust environments for businesses to sell products online. Similarly, many open-source projects and desktop applications demonstrate basic e-commerce functionality. However, most of these systems are either web-only, complex to configure, or not designed with educational clarity in mind. Some Java-based academic projects attempt to simulate small-scale shopping systems but often lack scalability, real-time database integration, and proper role-based access control.

Limitations of Current Systems

A common shortcoming in many of these systems is the absence of clear role-based separation. User and administrator functionalities are often blended together, leading to a cluttered user experience and increased security risk. Many JavaFX projects found in tutorials or student portfolios use tightly coupled UI and backend logic, making the codebase difficult to maintain or extend. Additionally, these systems frequently lack persistent storage using real databases, relying instead on text files or in-memory structures that do not reflect realistic application scenarios. Essential features like password recovery, session management, and data validation are frequently underdeveloped or entirely missing, making them impractical for real-world use or educational demonstrations.

How This Project Improves Upon Them

Tanzu's Creation addresses these issues by offering a clean, modular, and maintainable JavaFX-based desktop application with full role-based access. It separates customer and admin interfaces through distinct navigation paths and controllers, enhancing both usability and security. The application incorporates a complete shopping experience for users—including product browsing, cart management, checkout, and order tracking—while administrators are given dedicated tools for managing users, viewing orders, and controlling product inventory. Unlike many basic academic projects, Tanzu's Creation uses a real MySQL database for persistent storage, promoting practical understanding of database operations. By combining modern UI design, scalable architecture, and educational clarity, this project serves as both a functional e-commerce platform and a strong foundation for object-oriented software development practice.

7. System Analysis

Requirement Analysis

The primary goal of Tanzu's Creation is to build a user-friendly, desktop-based e-commerce platform that allows customers to browse products, manage a shopping cart, place orders, and track their order history, while giving administrators the ability to manage products, users, and orders through a dedicated panel. The system must provide secure login mechanisms, session handling, persistent storage, and intuitive navigation for both roles.

Functional Requirements

- User Registration and Login: Users should be able to register and securely log into the application.
- Role-Based Access Control: The system should differentiate between administrators and customers and load the appropriate interface based on role.
- Product Browsing: Customers should be able to view a list of available products with relevant details.
- Add to Cart: Users must be able to add products to a cart, view it, and update quantities.
- Checkout Process: Users should be able to place orders, view confirmation, and track their history.
- Admin Features:
 - Add, edit, and remove products
 - View all users
 - View all orders
- Password Recovery: Users should be able to reset forgotten passwords using email or recovery mechanisms.
- Session Handling: The system should maintain user sessions during navigation.

Non-Functional Requirements

- Usability: The UI should be intuitive and require minimal effort to navigate, even for non-technical users.
- Performance: The application should perform efficiently with minimal delays when switching views or querying the database.
- Maintainability: The system should follow the MVC pattern to ensure clean code structure and ease of future maintenance or expansion.
- Security: User credentials should be securely stored and validated, ensuring protected access to user and admin functionalities.
- Portability: As a JavaFX application, it should run on any platform with Java and MySQL installed.

Feasibility Study

Technical Feasibility

The application is built using JavaFX, FXML, and MySQL—all of which are well-supported, open-source technologies. JavaFX offers an accessible GUI framework, while FXML and CSS allow for clean UI separation. The system's use of a MySQL database ensures robust and scalable data management. These technologies are compatible and reliable for academic and small-business use.

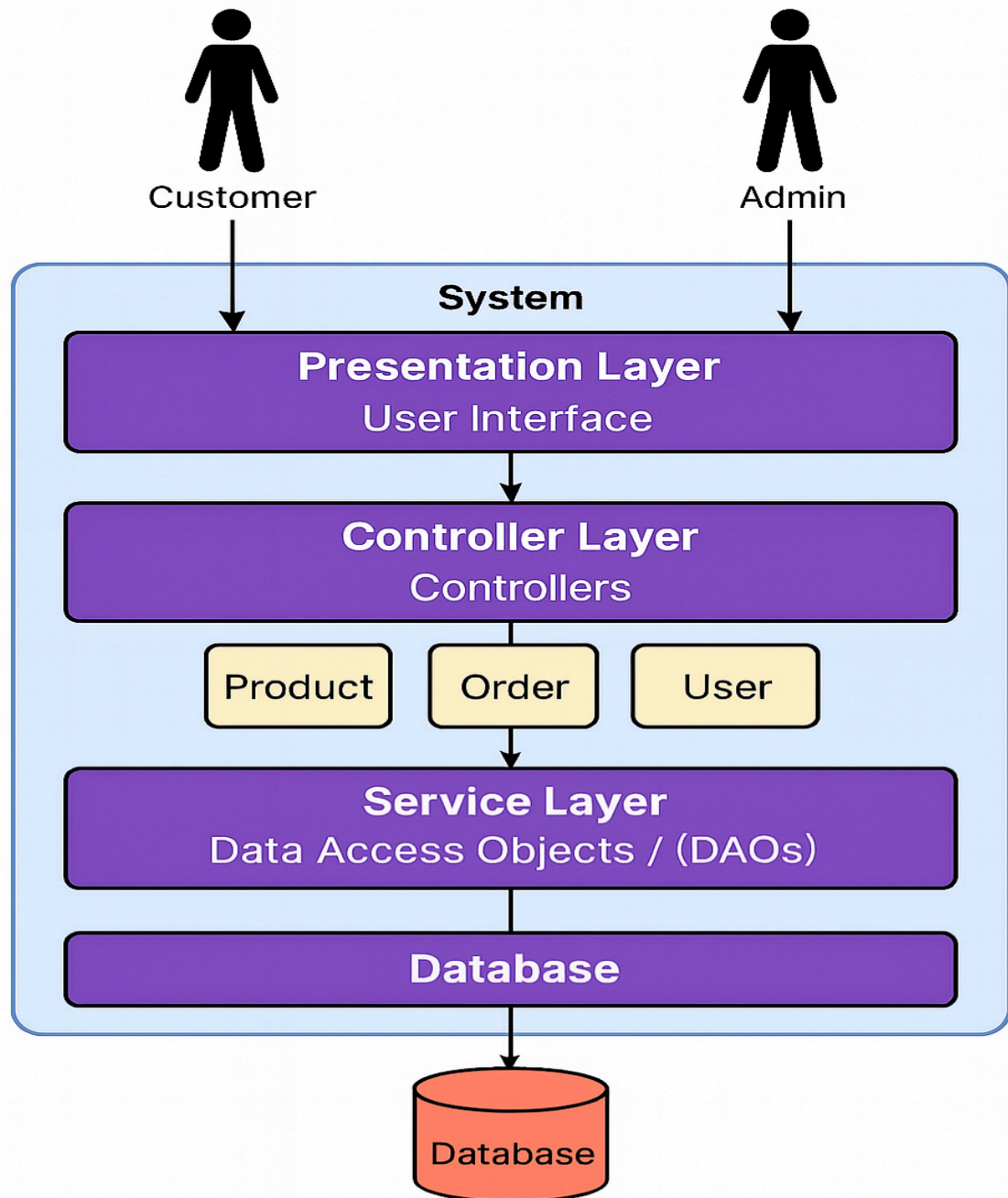
Economic Feasibility

This project does not require any proprietary software or third-party licensing. All technologies used are open-source, minimizing development costs. The only potential costs might involve hosting a database remotely or distributing the app with Java runtimes—both of which are optional. From an educational standpoint, the economic cost is negligible.

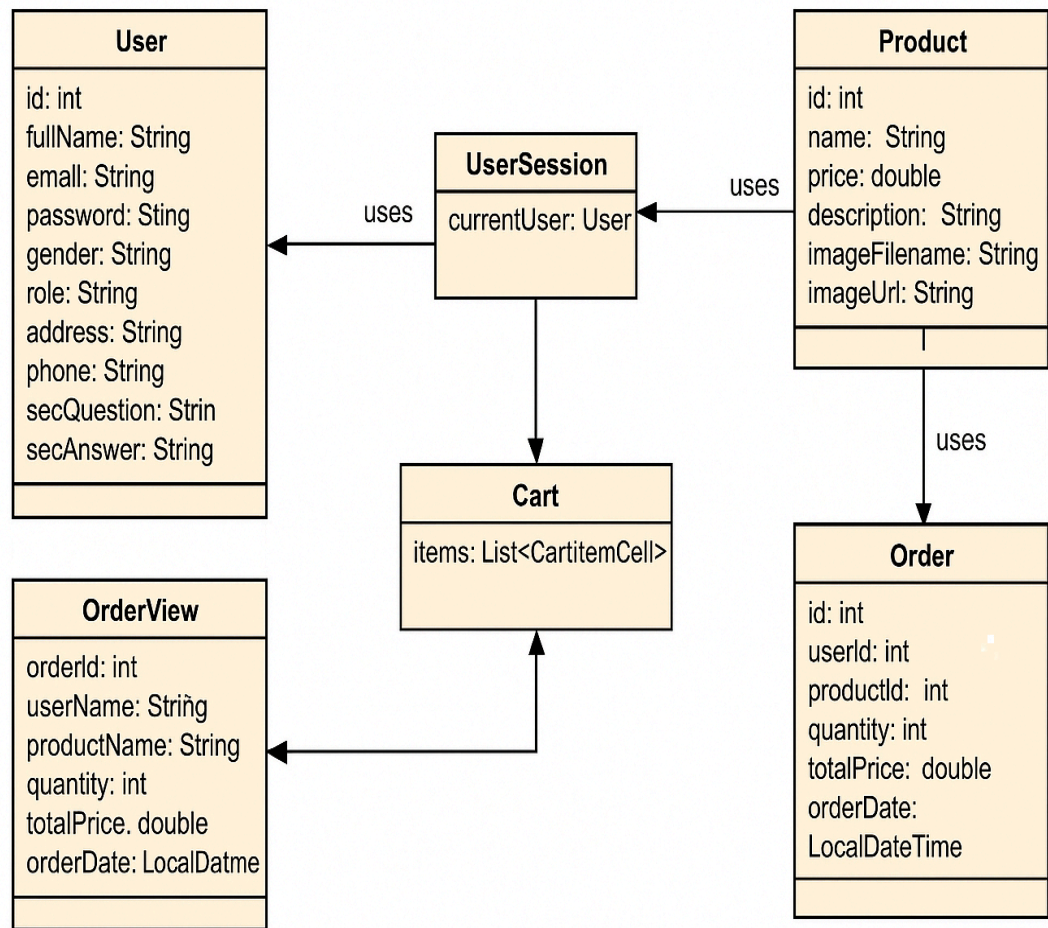
Operational Feasibility

The system is designed for easy operation by both technical and non-technical users. Customers can intuitively browse and shop, while administrators have simple controls for product and user management. The layout, session flow, and access roles ensure minimal training or onboarding is needed.

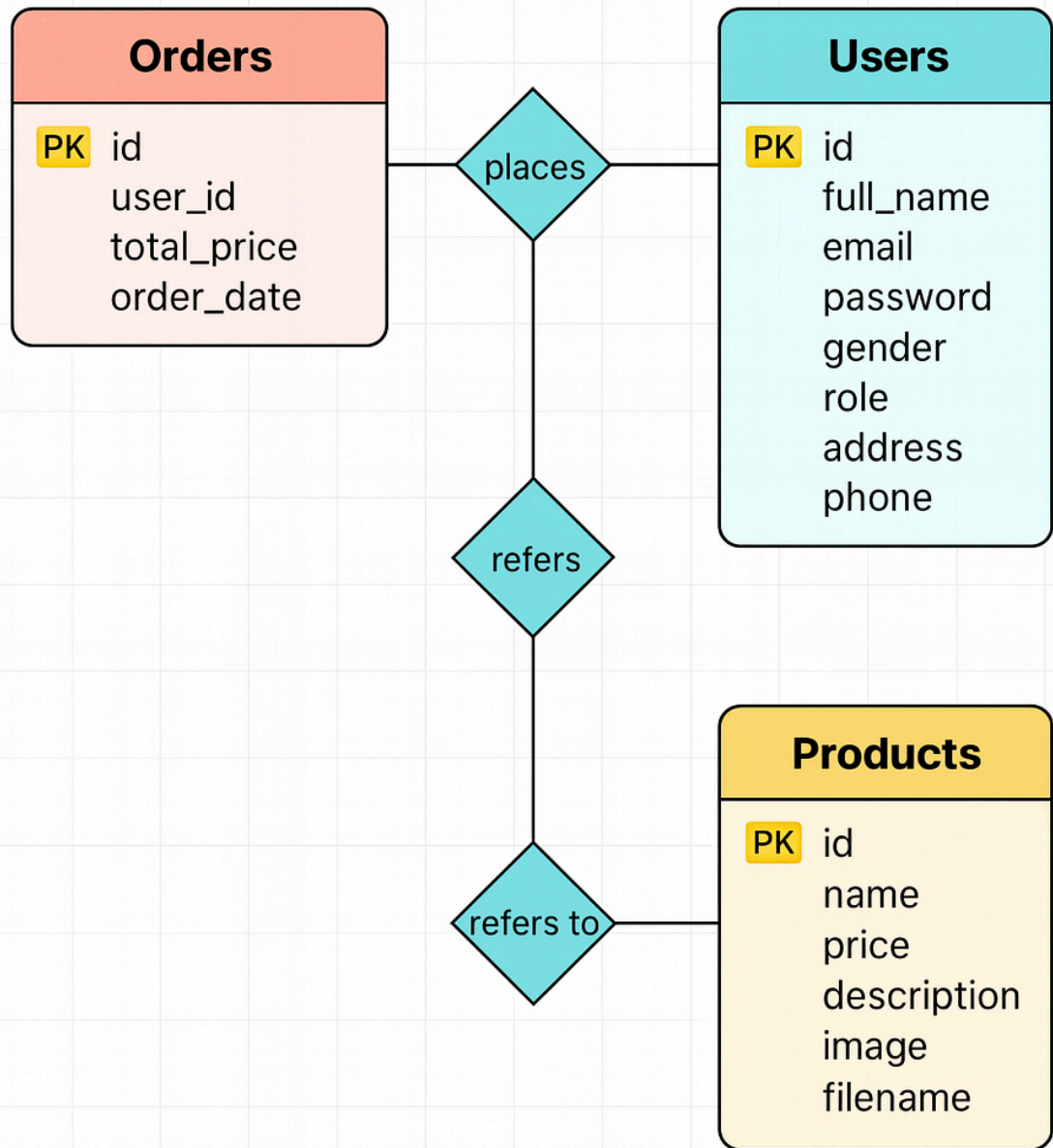
8. System Design



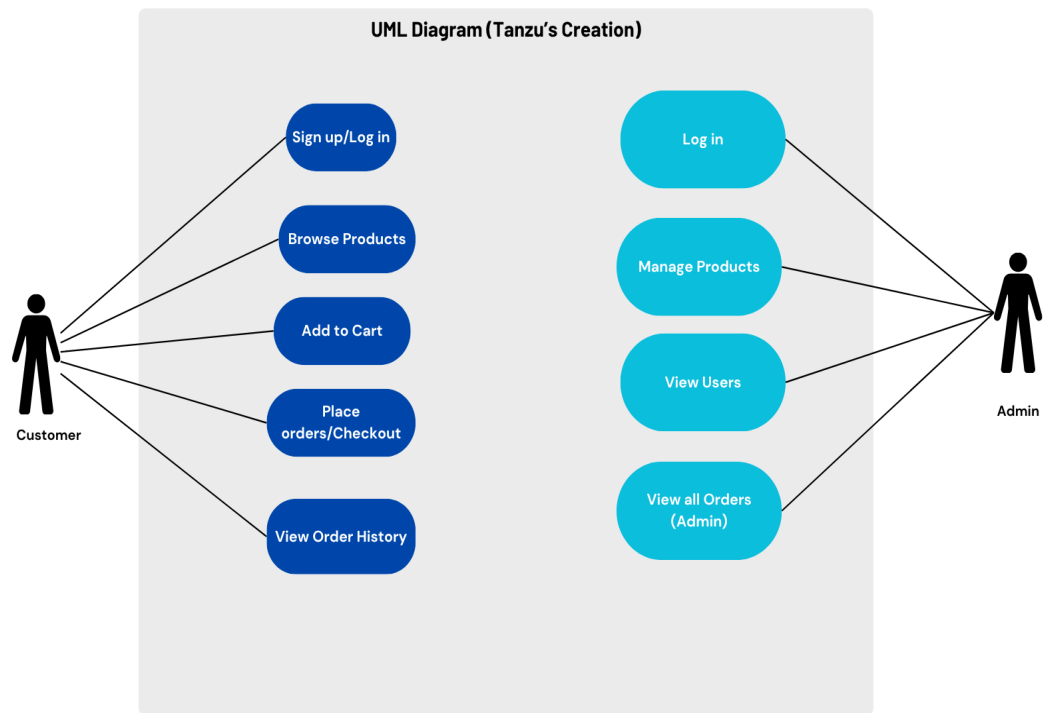
Tanzu's Creation Architecture Diagram



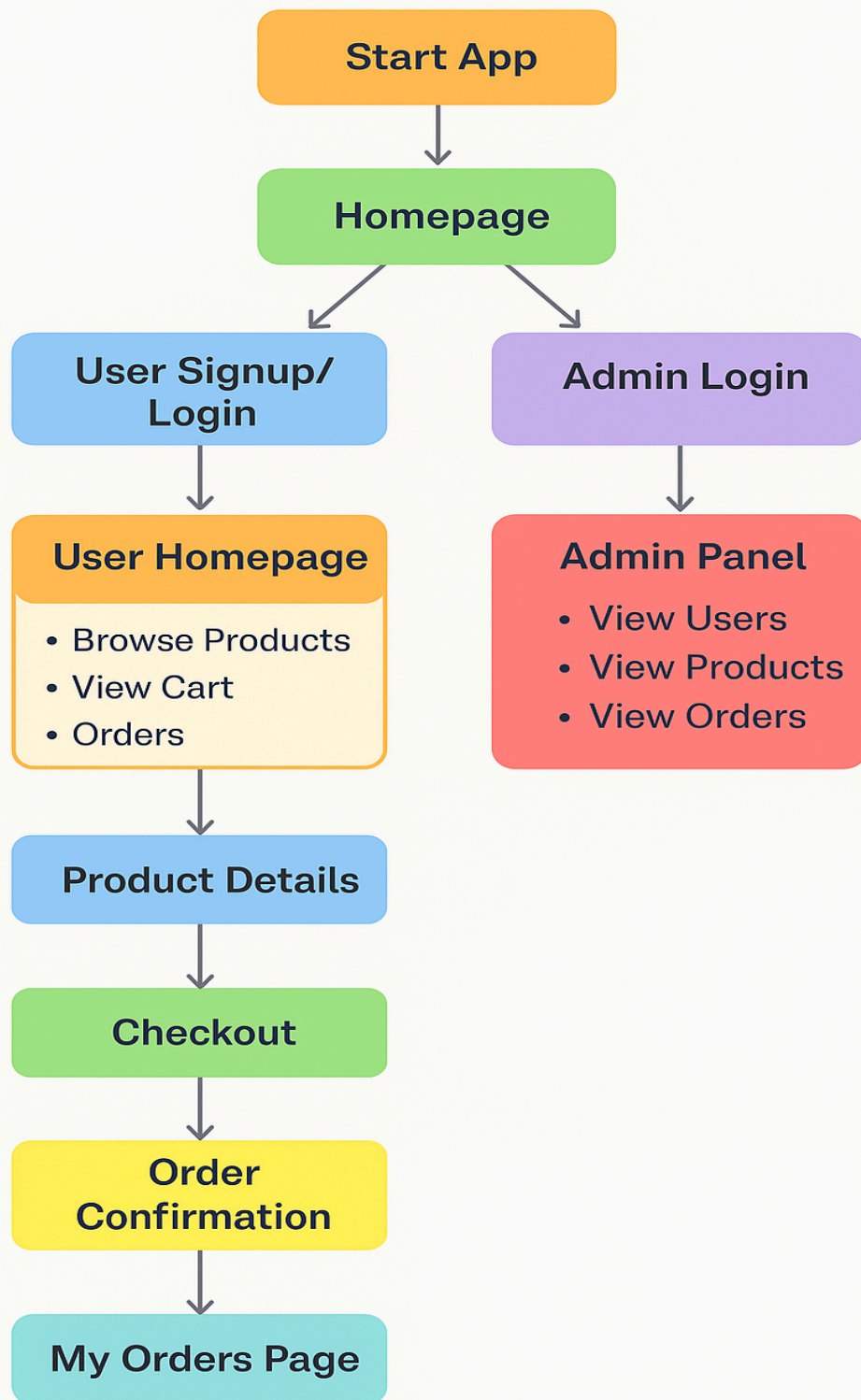
Tanzu's Creation-Class Diagram



Tanzu's Creation – ER Diagram



UML Diagram-Tanzu's Creation



Work flow diagram

User Interface Design explanation :

This document outlines the structural and stylistic design elements implemented across the user-facing and administrative interfaces of the application. Each screen is designed with consistency, usability, and an intuitive user experience in mind, leveraging JavaFX FXML layouts and custom CSS styling.

1. Start Screen / Homepage

- FXML File: `homepage.fxml`
- CSS File(s): `home.css`, `homepage.css`

Design Highlights:

- A visually appealing hero section with a banner (`.hero-section`) and a clear call-to-action button (`.hero-button`)
- Navigation buttons styled consistently using the `.nav-button` class
- Product previews displayed in tile format using:
 - `.product-card`
 - `.prod-title`
 - `.prod-price`
 - `.prod-cart-btn`
- The overall color scheme and background create a soft, friendly, pet-themed aesthetic to enhance user engagement

2. Login and Signup Screens

- FXML Files: `Login.fxml`, `signup.fxml`

- CSS Files: `login.css`, `signup.css`

Design Highlights:

- Centered forms placed over a themed background image (`images/bg.png`)
- Input fields styled with rounded corners using `.text-field` and `.password-field`
- Call-to-action buttons with gradient effects using `.button` and `.primary-button`
- User feedback elements styled with:
 - `.error-label` for validation messages
 - `.success-label` for positive confirmations

3. User Homepage (Post Login)

- FXML File: `homepage.fxml` (Dynamic Content Rendering)

Key Features:

- Product listings displayed using grid layout with image containers:
 - `.product-image-container`
 - `.product-image`
- Add to Cart functionality with buttons styled as `.add-to-cart-button`
- Scrollable interface implemented using `.scroll-pane` to manage content overflow

4. Product Detail Page

- FXML File: `product.fxml`
- Controller: `ProductController.java`

Design Highlights:

- Detailed presentation of an individual product including image, description, and pricing
- Action buttons for:
 - Adding the product to the cart
 - Continuing shopping

5. Cart Page

- FXML Files: `cart.fxml`, `CartItemCell.fxml`
- CSS File: `cart.css`

Design Highlights:

- Custom cell layout for each cart item
- Interactive controls for:
 - Adjusting quantity (`.quantity-button`)
 - Removing items from the cart (`.remove-button`)
- Total price summary displayed with emphasis using the `#checkoutButton`

6. Checkout Screen

- FXML File: `Checkout.fxml`
- CSS File: `checkout.css`

Design Highlights:

- A minimalist layout to confirm cart contents before placing an order
- Prominent green Checkout button styled as `.checkout-button`
- Total price highlighted in pink using the `#totalPriceLabel` selector

7. My Orders Page

- FXML File: `my_order.fxml`

Design Objective:

- A clean, structured list of the user's past orders
- Read-only display including:
 - Ordered items
 - Total cost
 - Order date and time

Admin Interface

Admin Dashboard

- FXML File: `admin_panel.fxml`
- CSS File: `admin_panel.css`

Design Highlights:

- Simplified dashboard layout with actionable sections:
 - Buttons styled as `.action-button`
 - Navigation options: View Users, View Products, View Orders

Admin Tables

- FXML Files: `view_users.fxml`, `view_products.fxml`, `view_orders.fxml`
- Controllers: Separate controller for each table view

Design Highlights:

- Use of `TableView` with clean borders and structured columns
- Data is read-only, aimed at administrative analysis and audit purposes
- Consistent styling for ease of data review

This comprehensive UI design ensures a seamless and visually consistent experience for both general users and administrators, supporting both usability and functionality across all application modules.

9. Implementation

Technologies Used

- **JavaFX:** The core framework for building the graphical user interface (GUI) with a modern desktop application feel.
- **FXML:** Used to design the UI layouts declaratively and separate them from business logic.
- **Scene Builder:** A drag-and-drop GUI tool for editing FXML files, streamlining interface design.
- **IDE:** Development was carried out using IDEs such as **Apache NetBeans** or **Eclipse**, providing support for JavaFX integration and debugging.
- **MySQL:** A relational database system used to persist user credentials, product details, cart items, and orders.
- **CSS:** Custom stylesheets were used to define the look and feel of the application interfaces.

Code Structure

The implementation of *Tanzu's Creation* follows a modular approach, leveraging the **Model-View-Controller (MVC)** architectural pattern to ensure separation of concerns, maintainability, and scalability. The codebase is organized into several logical packages and resource directories, each with a specific responsibility.

- **controllers package:** This package contains all JavaFX controller classes responsible for handling user interactions, managing scene transitions, and connecting the UI with the underlying logic. Examples include `LoginController.java`, `SignupController.java`, `HomepageController.java`, `CartController.java`, and `Admin_panelController.java`.
- **models package:** The models define the data structures and business logic of the application. This includes classes such as `User.java` for user data, `Product.java` for product details, `Cart.java` and `Order.java` for shopping and transaction management, and `UserSession.java` for maintaining session state throughout the application.
- **fxml directory:** This directory contains all user interface layout files designed using FXML. These files define the visual structure of the application screens, including `Login.fxml`, `homepage.fxml`, `cart.fxml`, `Checkout.fxml`, and `admin_panel.fxml`.

- **css directory:** The CSS directory includes styling files used to define the appearance and theming of each interface. Each major scene has its own dedicated stylesheet, such as `login.css`, `signup.css`, `cart.css`, `checkout.css`, and `admin_panel.css`, ensuring a consistent and responsive user experience.
- **Main.java:** This serves as the entry point of the application. It initializes the JavaFX runtime and loads the initial login scene.

This organized structure not only supports efficient development but also aligns with best practices for JavaFX application design. It allows for easy navigation, modular debugging, and the potential for future extension or refactoring.

Modules and Packages

- **Authentication:** `LoginController`, `SignupController`, `Forgot_passwordController`
- **Customer Interface:** `HomepageController`, `CartController`, `CheckoutController`, `My_orderController`
- **Admin Interface:** `Admin_panelController`, `View_usersController`, `view_productsController`, `View_ordersController`
- **Shared Models:** `User`, `Product`, `Cart`, `Order`, `UserSession`

Explanation of Major Classes and Methods

- **Main.java:** The entry point of the application. It loads the login screen initially.
- **UserSession.java:** Stores currently logged-in user info and role. Used across controllers for session persistence.

- **ProductController.java**: Handles logic for loading product details, displaying them dynamically, and allowing interaction such as adding to cart.
- **CartController.java**: Manages items in the shopping cart. Handles quantity updates, removals, and checkout transitions.
- **Admin_panelController.java**: Central controller for admin features like navigating to user/product/order views.
- **OrderView.java / Order.java**: Models used to manage and display order details for both users and admins.

FXML & Controller Interaction

- Each **.fxml** file defines the layout for a specific screen using declarative XML-style syntax.
- Controllers (e.g., **LoginController**) are bound to these FXML files using the **fx:controller** attribute.
- Within each controller:
 - UI elements are injected using **@FXML** annotations.
 - Action handlers (e.g., **onLoginButtonClick**) are linked via the FXML **onAction** property.
 - Scene switching is handled using **FXMLLoader.load()** and **setScene()** in **Stage**.

Example:

@FXML

```
private void handleLogin(ActionEvent event) {

    if (authenticateUser(username.getText(), password.getText())) {

        FXMLLoader loader = new FXMLLoader(getClass().getResource("/fxml/homepage.fxml"));

        Parent root = loader.load();
```

```
stage.setScene(new Scene(root));  
  
stage.show();  
  
}  
  
}
```

This separation ensures that **logic stays in Java**, while **design stays in FXML**, following best practices.

10. Testing

The testing phase of *Tanzu's Creation* involved a systematic evaluation of all functional components to ensure the application behaves as intended. Testing focused on validating both typical user interactions and edge cases, using manual walkthroughs and scenario-based input validation. The goal was to confirm correctness, usability, and role-based flow integrity across both customer and administrator functionalities.

Test Cases and Outcomes

Testing covered all major features of the application. For each feature, the system was evaluated based on expected input and corresponding output behavior:

- **User registration and login functionality** was tested by entering valid and invalid credentials. The application correctly differentiated between registered and unregistered users, and directed each role to the appropriate interface upon successful authentication.
- **Product browsing and cart management** features were tested by simulating a standard shopping flow. Users could add products to the cart, modify quantities, remove items, and proceed to checkout. The system updated totals dynamically and responded accurately to user input.
- **Order placement and tracking** were tested by completing a checkout process and then verifying that the order was correctly recorded and viewable in the user's order history. Data consistency between the cart and order database was maintained.
- **Administrative features**, including viewing products, users, and orders, were tested for data accuracy and restricted access. The admin panel correctly loaded datasets from the database and

prevented regular users from accessing protected views.

- **Password recovery mechanisms** were validated to ensure that invalid or blank email entries produced appropriate error messages, and that valid inputs triggered the expected recovery process.
- **Form validation and input handling** were tested across all relevant scenes. Fields that were left empty, incorrectly formatted, or inconsistent with backend validation rules were appropriately flagged, and the system prevented continuation until corrected.

All tested scenarios yielded expected results, and the interface provided immediate, user-friendly feedback when issues were detected.

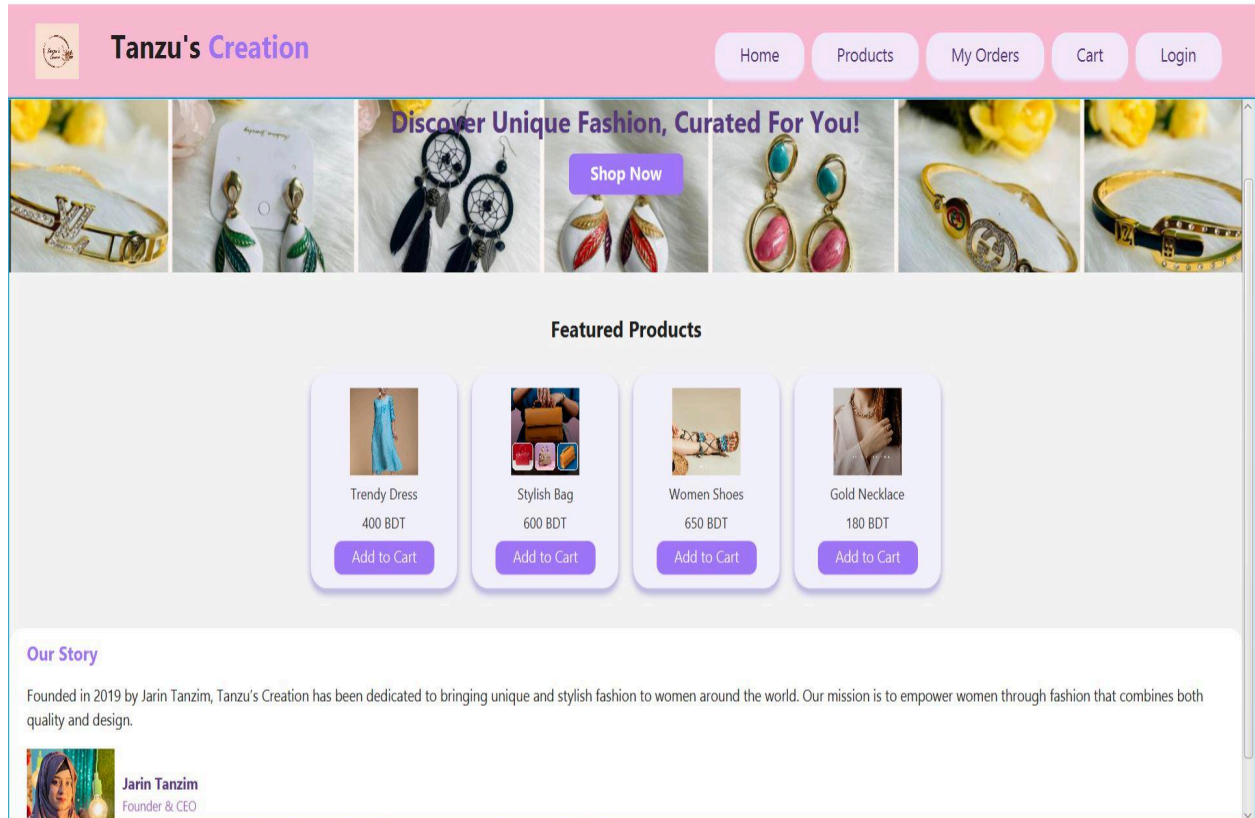
Identified Issues and Resolutions

Several issues emerged during early stages of testing:

- In some instances, cart items were lost during scene transitions. This was resolved by implementing a centralized session management class (`UserSession.java`) to persist state across views.
- The password recovery form initially accepted blank inputs without prompt. Validation was added to enforce email formatting and required input before submission.
- Administrative views were accessible from user accounts due to missing role checks. This was resolved by incorporating explicit role-based redirection logic during the login process.
- Rapid scene transitions led to UI flickering on lower-performance systems. This was addressed by implementing preloading strategies for common scenes.

All identified bugs were addressed and re-tested to confirm that the fixes were effective and did not introduce new issues.

11. Screenshots / UI Snapshots



Sign Up

Back



Create your Account

Select your Gender

☐ Male ☐ Female

Security Question:

Sign up

Back



Log in to Tanzu's Creation

Log in

Forgot Password?

Don't have an account? [Register here](#)

Forgot Password

Enter your registered email:

Next

Security Question: What is your favorite food?

Verify

[Back to Login](#)

Forgot Password

Enter your registered email:

Next

Enter new password:

Reset Password

[Back to Login](#)



Products

[Home](#)
[Cart](#)
[My orders](#)
[Log in](#)



Bracelet
350 BDT
[Add to Cart](#)



Earring
350 BDT
[Add to Cart](#)



Earring
350 BDT
[Add to Cart](#)



Earring
350 BDT
[Add to Cart](#)



Dress
980 BDT
[Add to Cart](#)



Bracelet
350 BDT
[Add to Cart](#)



Earring
350 BDT
[Add to Cart](#)



Bracelet
350 BDT
[Add to Cart](#)









Your Cart

	Women Shoes	-	1	+	650.00 B...	X
	Trendy Dress	-	2	+	800.00 B...	X
						

[Continue Shopping](#)

Total items: 4
Total price: 1630.00 ...
[Checkout](#)

[Back](#)

Checkout

Shipping Info

Jarin

Order Summary

Total: **1630.00 BDT**

Payment Method

bKash

Confirm Order

[Back](#)

Checkout

Shipping Info

Jarin

Order Summary

Total: **1630.00 BDT**

Order Confirmed



Your order has been placed successfully!

OK

My Orders

Image	Product	Quantity	Total Price	
	Women Shoes	1	650.0	
	Braclet	1	350.0	
	Braclet	1	350.0	
	Gold Necklace	1	180.0	
	Braclet	2	700.0	
	Stylish Bag	1	600.0	

[Back](#)

Admin Dashboard

[Log Out](#)

Welcome, Admin!

 [View Users](#)

 [Add Products](#)

 [View Products](#)

 [View Orders](#)

[illegible]

Add New Product

Add New Product

Earring

220

Sea.jpg

Handmade Earring

Add Product

Back

Product List

ID	Name	Price	Image	Action		
7	Earring	195.0	Ear 15.jpg	Edit	Delete	
9	Ear 2	100.0	Ear 2.jpg	Edit	Delete	
10	Kundan	450.0	kundan1.jpg	Edit	Delete	
11	Bangles	340.0	churi.jpg	Edit	Delete	
12	Braclet 1	300.0	Braclet 1.jpg	Edit	Delete	
15	Earring	210.0	Ear 13.jpg	Edit	Delete	
16	Earring	320.0	Ear 14.jpg	Edit	Delete	
17	Braclet	350.0	Braclet 4.jpg	Edit	Delete	
18	Earring	350.0	Ear 7.jpg	Edit	Delete	
19	Earring	350.0	Ear 12.jpg	Edit	Delete	
20	Earring	350.0	Ear 1.jpg	Edit	Delete	
21	Dress	980.0	dress.png	Edit	Delete	
22	Braclet	350.0	Braclet 1.jpg	Edit	Delete	
23	Earring	350.0	Ear 9.jpg	Edit	Delete	
24	Braclet	350.0	braclet 2.jpg	Edit	Delete	

[Back to Admin Panel](#)

Order List

Order ID	User	Product	Quantity	Total Price	
1	11	3	1	350.0	
2	1	3	1	350.0	
3	11	3	1	195.0	
4	11	29	1	600.0	
5	11	3	1	350.0	
6	11	30	2	1300.0	
7	11	3	1	350.0	
8	11	21	1	980.0	
9	1	30	1	650.0	
10	1	17	1	350.0	
11	1	17	1	350.0	
12	1	3	1	350.0	
13	1	3	2	700.0	
14	1	31	1	180.0	
15	1	17	2	700.0	
16	1	29	1	600.0	
17	1	21	1	980.0	
18	1	29	1	600.0	
19	1	30	1	650.0	
20	1	31	1	180.0	

[Back to Admin Panel](#)

12. Conclusion

Summary of Work Done

Tanzu's Creation is a JavaFX-based desktop e-commerce application developed as part of the course CSE-3170: Object Oriented Software Development Lab. The primary goal of the project was to design and implement a complete system that enables users to browse products, manage a shopping cart, place orders, and track their purchase history, while also equipping administrators with tools to manage products, users, and orders. The application was built using a modular, MVC-based architecture, incorporating FXML for layout design, CSS for UI styling, and MySQL for persistent data management. Both customer and admin roles were supported through secure login and session management, and the system was thoroughly tested to ensure reliable functionality.

Challenges Faced and How They Were Overcome

One of the key challenges during development was maintaining state across multiple views in a desktop-based system. This issue was particularly significant for cart management and session handling. To address this, a custom `UserSession` class was implemented to store user-specific data persistently during runtime.

Another major challenge was implementing clear role-based access control. Initially, users and admins could access each other's views without restriction. This was resolved by introducing logic that verifies user roles upon login and conditionally loads the appropriate interface.

Designing an intuitive and visually appealing user interface also presented challenges, especially in coordinating FXML layout structure with CSS-based styling. This was mitigated by using Scene Builder for layout design and carefully organizing stylesheet rules for consistency.

Finally, database integration required careful handling of queries and data synchronization, particularly to ensure consistency between cart contents and order records. Structured SQL queries and input validation checks were used to maintain data integrity.

Achievements

The final implementation of *Tanzu's Creation* successfully meets all specified functional and non-functional requirements. It features secure user authentication, a clean and responsive user interface, full shopping cart and checkout functionality, and a robust admin panel for monitoring and managing platform activity.

From a technical perspective, the project demonstrates strong understanding and application of object-oriented principles, GUI design, and persistent data management using JavaFX and MySQL. It serves as a fully functional, maintainable, and extendable application that reflects industry-standard architectural practices. Most importantly, it offers a practical solution for small-scale digital commerce while serving as a valuable academic exercise in software development.

13. Future Work

While *Tanzu's Creation* meets the core functional and non-functional requirements of a desktop-based e-commerce system, several areas offer scope for future development and enhancement. These potential extensions could improve usability, broaden the user base, and align the application more closely with modern commercial platforms.

Features That Can Be Added

To expand the application's functionality, several advanced features could be introduced. These include:

- **Product Search and Filtering:** Implementing a dynamic search bar and category-based filters would enhance user navigation and improve product discoverability.
- **User Profile Management:** Allowing users to edit personal information, change passwords, and manage delivery preferences could personalize the experience.
- **Order Status Tracking:** Integrating order status updates (e.g., pending, shipped, delivered) would improve transparency and trust for customers.
- **Notifications:** Adding in-app notifications or email alerts for order confirmation, delivery updates, or administrative messages would enhance user engagement.

Migration to Mobile or Web

Currently, the application is built as a desktop system using JavaFX. However, modern users increasingly prefer mobile and web-based platforms. Future development could focus on:

- **Web Migration:** Rebuilding the system using web frameworks such as React (frontend) and Spring Boot or Django (backend) would make it accessible via browsers and support wider deployment.
- **Mobile App Integration:** Using tools like Flutter or React Native, a mobile version could be developed that connects to the same backend, offering a seamless multi-platform experience.
- **API-Driven Architecture:** Introducing RESTful APIs would decouple the frontend and backend, enabling integration with other systems or third-party services.

Enhancements in UI/UX or Scalability

As user expectations continue to evolve, so should the design and architecture of the system:

- **Modernized UI Design:** Transitioning to a more responsive and material-design-inspired layout could make the interface more appealing and intuitive.
- **Dark Mode and Accessibility Features:** Including theme customization and accessibility support (e.g., screen reader compatibility, larger fonts) would broaden usability.
- **Database Optimization:** As data volume grows, optimizing SQL queries and incorporating indexing or caching mechanisms will improve performance.
- **Multi-Tenant Support:** For broader commercial use, the application could support multiple vendors or administrators with isolated dashboards and analytics.

These proposed enhancements not only align the application with current trends in software development but also provide a clear roadmap for continued growth, scalability, and cross-platform reach. Implementing them would elevate *Tanzu's Creation* from an academic prototype to a production-ready digital commerce solution.

14. References

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15. Appendix

A. Source Code Repository

The full source code for *Tanzu's Creation* is available upon request or can be accessed via a version-controlled repository. The repository includes all Java source files, FXML layouts, CSS stylesheets, and database scripts required to run the application.

GitHub Repository: <https://github.com/Jarin-Tanzim/Tanzu-s-Creation-Desktop-App-JavaFX->